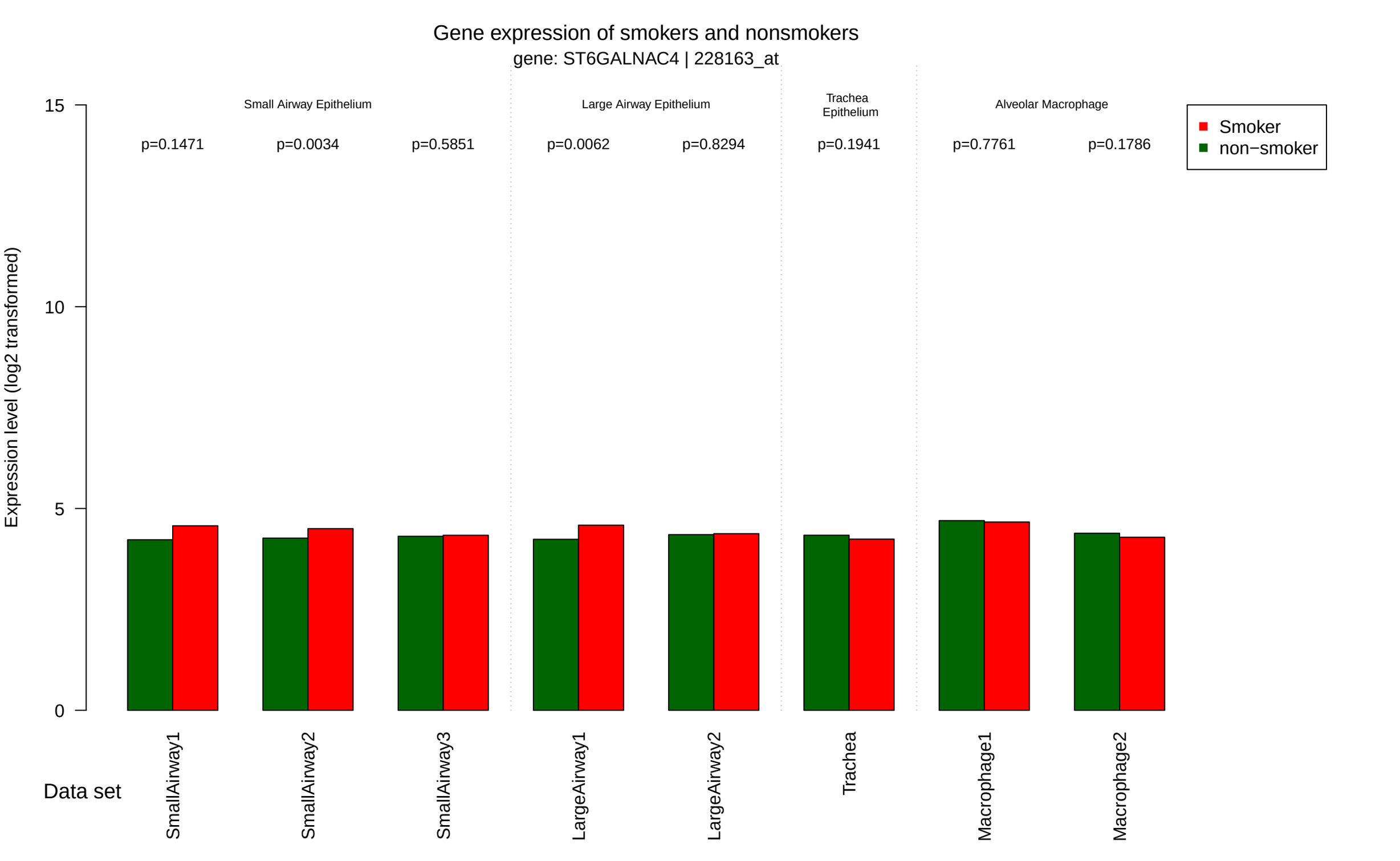
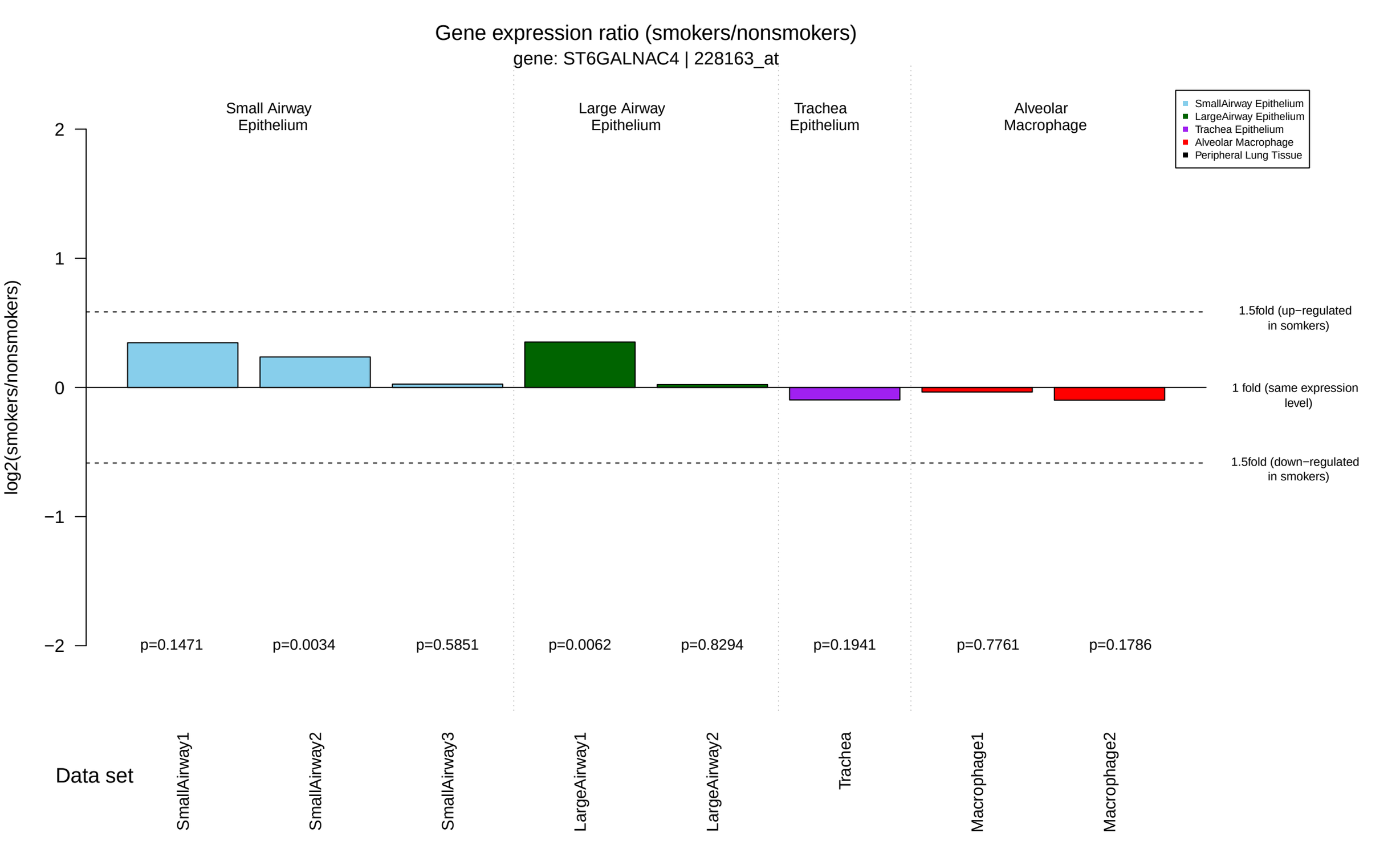
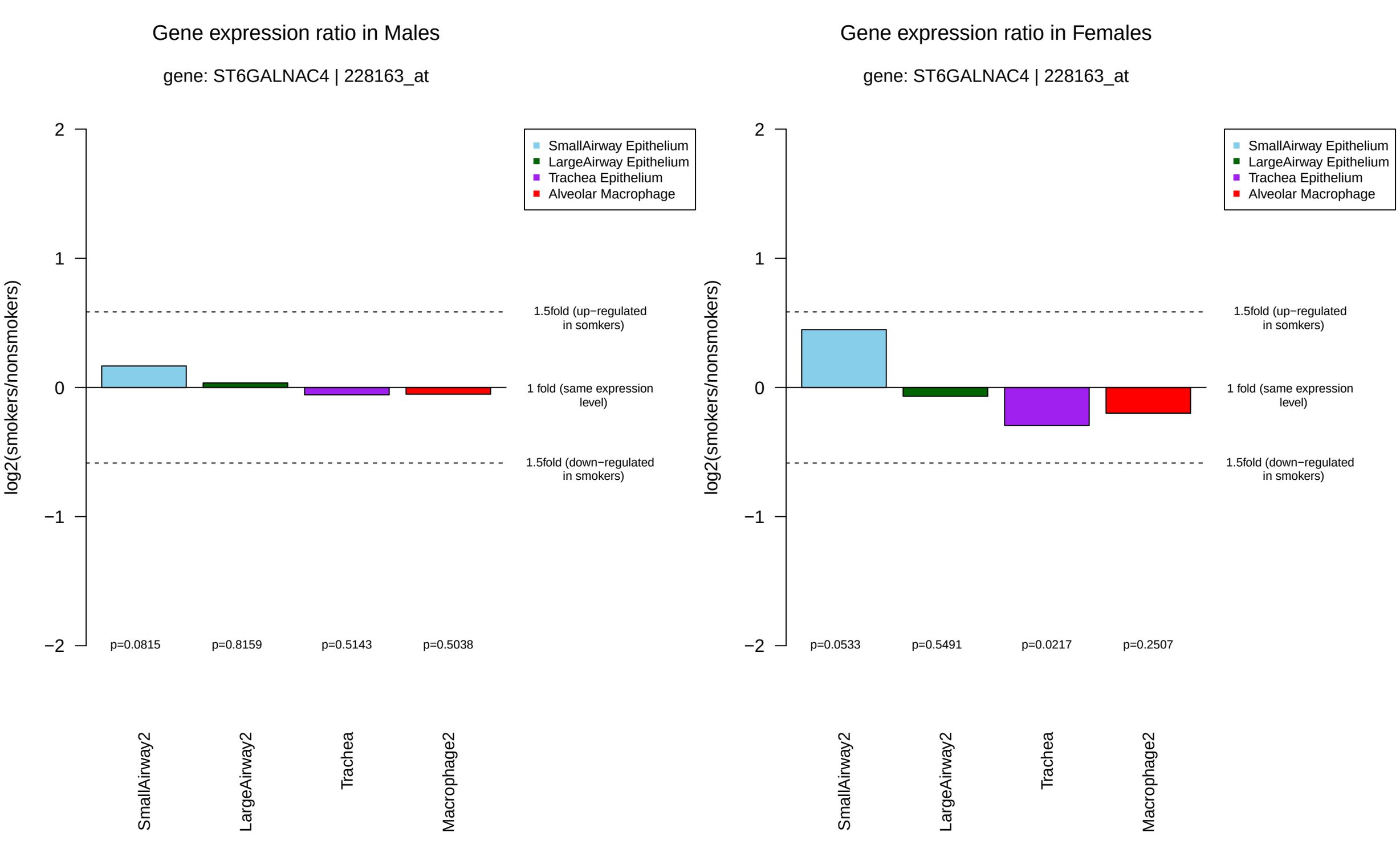
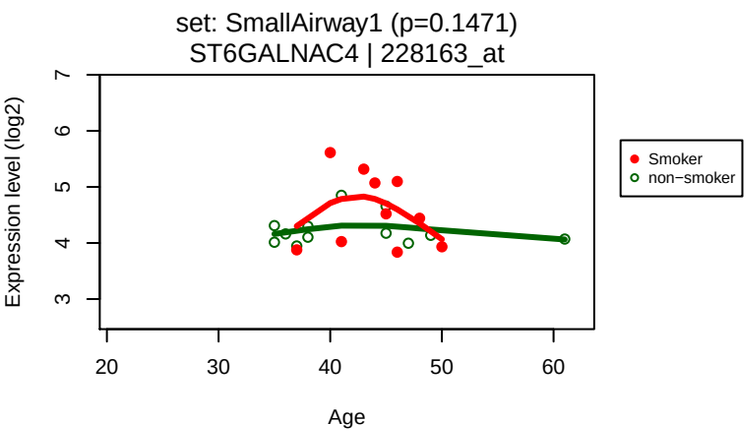
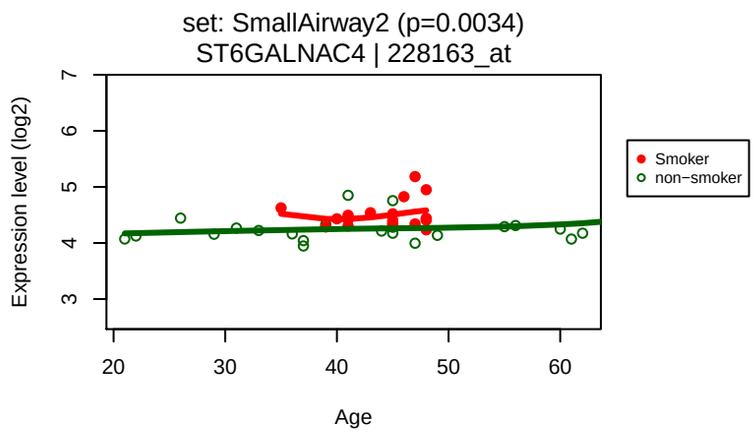
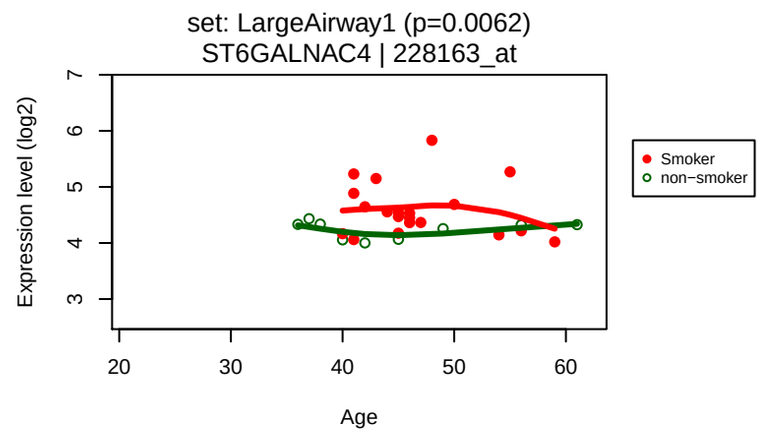

Gene expression of smokers and nonsmokers
gene: ST6GALNAC4 | 228163_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1471
p=0.0034
p=0.5851
p=0.0062
p=0.8294
p=0.1941
p=0.7761
p=0.1786
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

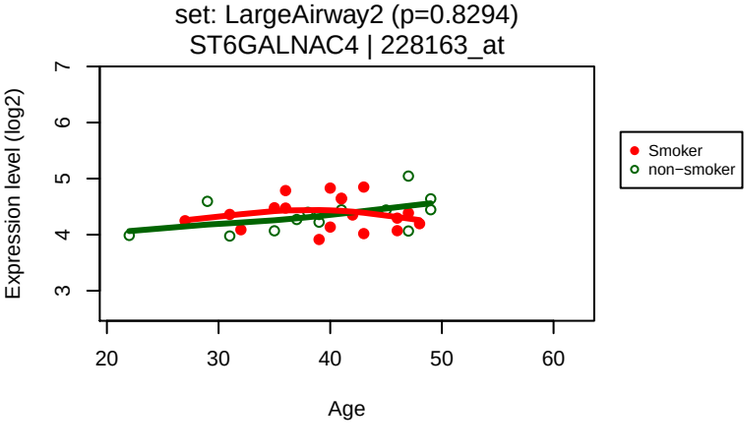

Gene expression ratio (smokers/nonsmokers)
gene: ST6GALNAC4 | 228163_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1471
p=0.0034
p=0.5851
p=0.0062
p=0.8294
p=0.1941
p=0.7761
p=0.1786
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

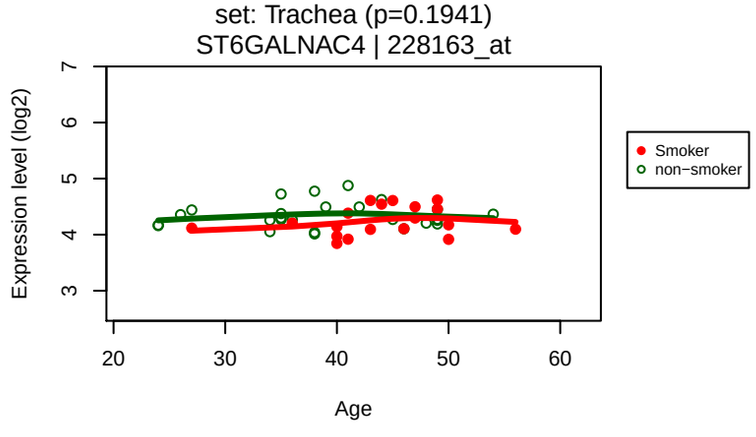

Gene expression ratio in Males
gene: ST6GALNAC4 | 228163_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0815
p=0.8159
p=0.5143
p=0.5038
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ST6GALNAC4 | 228163_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0533
p=0.5491
p=0.0217
p=0.2507
SmallAirway2
LargeAirway2
Trachea
Macrophage2

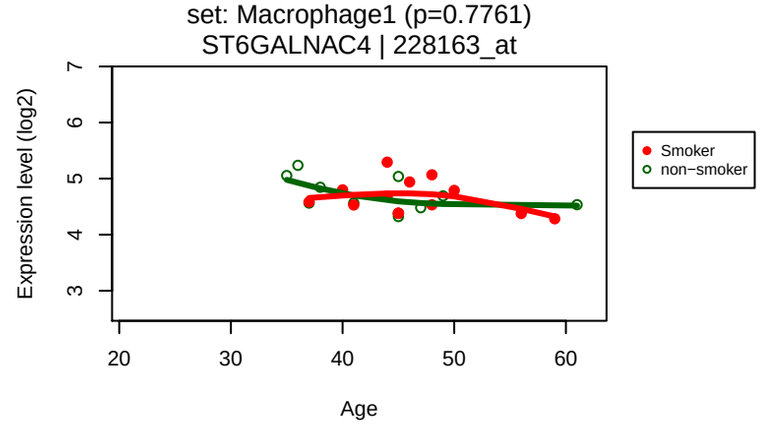

set: SmallAirway1 (p=0.1471)
ST6GALNAC4 | 228163_at
Expression level (log2)
Age
Smoker
non−smoker

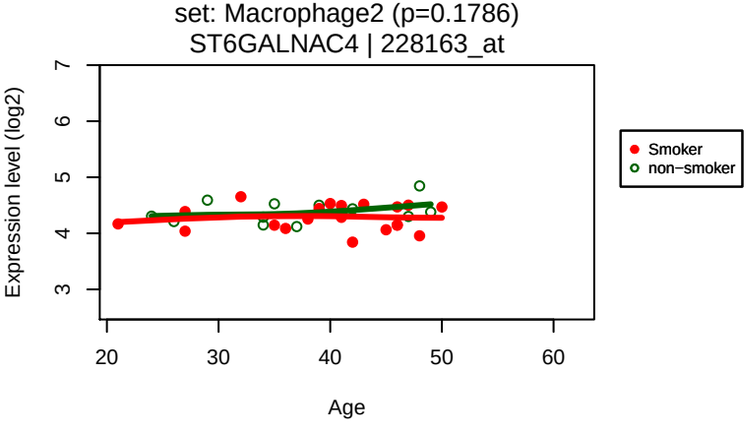

set: SmallAirway2 (p=0.0034)
ST6GALNAC4 | 228163_at
Expression level (log2)
Age
Smoker
non−smoker

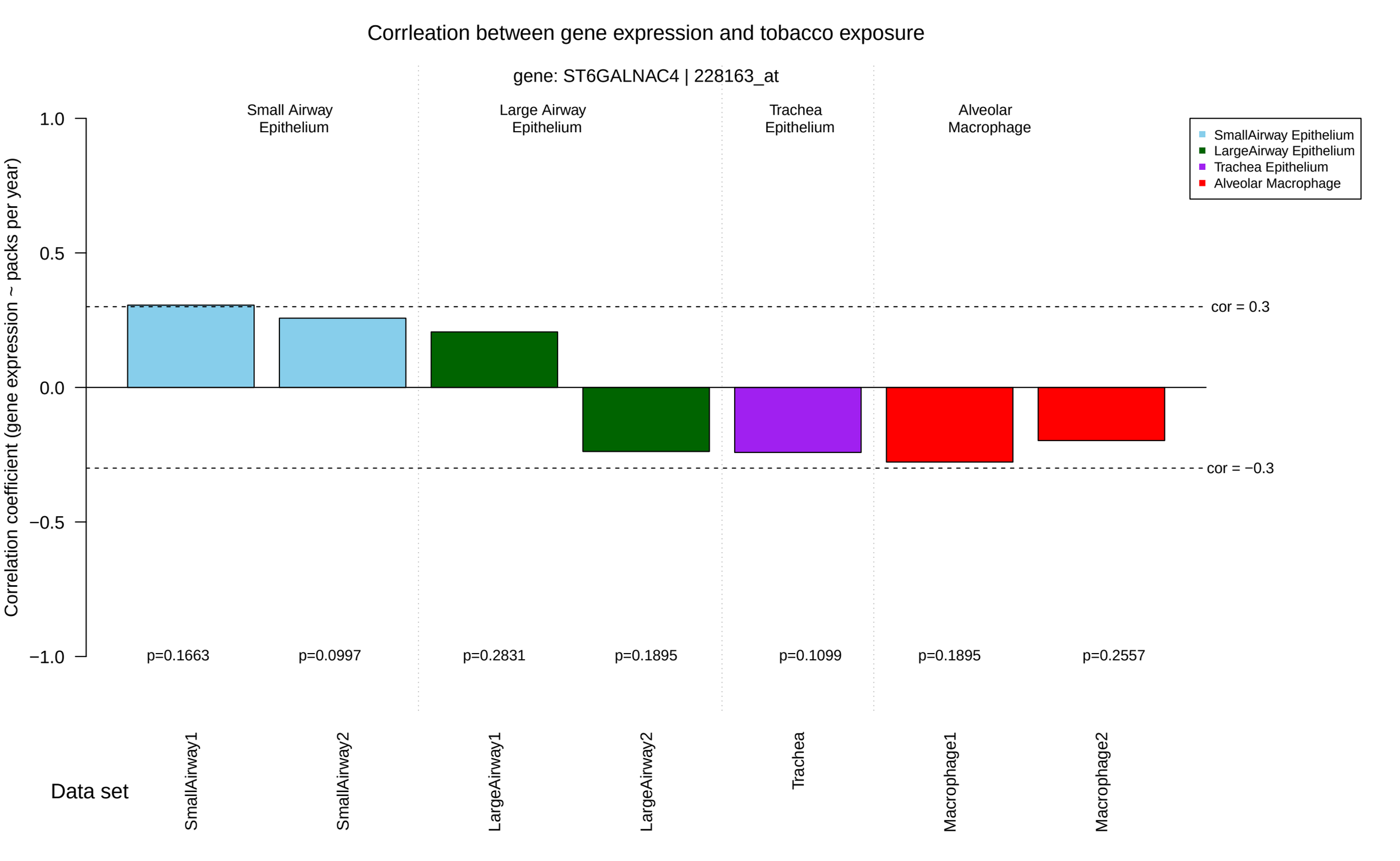

set: LargeAirway1 (p=0.0062)
ST6GALNAC4 | 228163_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8294)
ST6GALNAC4 | 228163_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1941)
ST6GALNAC4 | 228163_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7761)
ST6GALNAC4 | 228163_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1786)
ST6GALNAC4 | 228163_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ST6GALNAC4 | 228163_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1663
p=0.0997
p=0.2831
p=0.1895
p=0.1099
p=0.1895
p=0.2557
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2